

ECON602 TA Notes: Hsieh and Klenow (2009)

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This document presents the details of Hsieh and Klenow (2009, QJE). It is one of the most cited QJE papers of the past decade and has generated a large subsequent literature on misallocation. However, the paper contains several errors: roughly one-quarter of its equations are not exactly correct. The main objective of this document is to clarify these issues.

1 Theoretical Framework

There are three layers of production in the model: final good Y , sectoral good Y_s , and differentiated good Y_{si} .

1.1 Layers of Firms

Final Good Producer

Final good Y is competitively produced from sectoral goods Y_s :

$$Y = \prod_s Y_s^{\theta_s} \quad (1)$$

where $\sum_s \theta_s = 1$, implying constant returns to scale.

Given the final-good price P and sectoral-good prices P_s , the final-good producer solves

$$\max_{Y_s} PY - \sum_s P_s Y_s$$

subject to (1). FOC is therefore

$$\theta_s PY = P_s Y_s \quad (2)$$

Sectoral Good Producer

Each sectoral good Y_s is produced from differentiated goods Y_{si} , using a CES production technology:

$$Y_s = \left(\sum_{i=1}^{M_s} Y_{si}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}} \quad (3)$$

The sectoral good producer solves

$$\max_{Y_{si}} P_s Y_s - \sum_{i=1}^{M_s} P_{si} Y_{si}$$

subject to (3). FOC gives the demand for differentiated good Y_{si} :

$$P_s Y_s^{1/\sigma} = P_{si} Y_{si}^{1/\sigma} \quad (4)$$

Differentiated Good Producer

The differentiated-good producer chooses output Y_{si} , price P_{si} , capital input K_{si} , and labor input L_{si} to maximize profit:

$$(1 - \tau_{Y_{si}}) P_{si} Y_{si} - (1 + \tau_{K_{si}}) R K_{si} - w L_{si}$$

subject to the demand function (4) and the production function

$$A_{si} K_{si}^{\alpha_s} L_{si}^{1-\alpha_s} \geq Y_{si} \quad (5)$$

For now, interpret τ_{Ksi} and τ_{Ysi} as taxes imposed on the capital acquisition and revenue of firm i in industry s .

As usual, the optimization can be carried out in two stages. In the first stage, for any given output Y_{si} , the differentiated good producer solves the cost minimization problem

$$\min_{K_{si}, L_{si}} (1 + \tau_{Ksi})RK_{si} + wL_{si}$$

subject to the production function (5).

Combining the FOCs for capital and labor gives the standard “MRTS = price” condition:

$$\frac{(1 + \tau_{Ksi})R}{w} = \frac{\alpha_s K_{si}^{\alpha_s - 1} L_{si}^{1 - \alpha_s}}{(1 - \alpha_s) K_{si}^{\alpha_s} L_{si}^{-\alpha_s}} = \frac{\alpha_s}{1 - \alpha_s} \frac{L_{si}}{K_{si}} \quad (6)$$

and marginal cost is simply the Lagrange multiplier of this cost minimization problem

$$\lambda_{si} = \frac{(1 + \tau_{Ksi})R}{\alpha_s A_{si} K_{si}^{\alpha_s - 1} L_{si}^{1 - \alpha_s}} = \frac{w}{(1 - \alpha_s) A_{si} K_{si}^{\alpha_s} L_{si}^{-\alpha_s}} \quad (7)$$

The cost minimization problem above implies a cost function $\mathcal{C}(Y_{si})$, with $\mathcal{C}'(Y_{si}) = \lambda_{si}$.

Note:

- τ_{Ksi} is akin to a tax on capital acquisition. More generally, it is a “wedge” in the first-order condition for input acquisition (6): it prevents the marginal rate of technological substitution $\frac{\alpha_s}{1 - \alpha_s} \frac{L_{si}}{K_{si}}$ from equaling the relative input price $\frac{R}{w}$.
- In other words, any mechanism that prevents the MRTS from equaling the relative price contributes to the wedge τ_{Ksi} . For instance, the firm may face financial frictions that restrict capital adjustment or labor-adjustment costs associated with hiring and layoffs. Such mechanisms can generate variation in τ_{Ksi} both in the cross-section and over time. The wedge τ_{Ksi} is a reduced-form representation of these mechanisms.
- Wedges are useful diagnostic tools for identifying the dominant frictions that prevent the MRTS from equaling the relative price. For instance, if τ_{Ksi} is countercyclical in the data, mechanisms that generate countercyclical distortions may dominate those that generate procyclical distortions. For an example, see Chari, Kehoe, and McGrattan’s “Business Cycle Accounting” paper.
- Also, note that different firms face different τ_{Ksi} . Hence, these idiosyncratic distortions are an important source of heterogeneity.

- Finally, the paper is not explicit about the relationship between $\tau_{K_{si}}$ and A_{si} . In the literature, a distortion correlated with productivity is called a “correlated distortion.” Antitrust policy is one example: large, highly productive firms may face higher taxes. Correlated and uncorrelated distortions have different implications for firm dynamics (see the discussion in ECON 701).

Now I go to the second stage of the firm’s problem where prices and quantities are chosen:

$$\max_{P_{si}, Y_{si}} (1 - \tau_{Y_{si}}) P_{si} Y_{si} - \mathcal{C}(Y_{si})$$

subject to (4).

FOC:

$$(1 - \tau_{Y_{si}}) P_{si} = \frac{\sigma}{\sigma - 1} \lambda_{si} = \frac{\sigma}{\sigma - 1} \frac{(1 + \tau_{K_{si}}) R}{\alpha_s A_{si} K_{si}^{\alpha_s - 1} L_{si}^{1 - \alpha_s}} = \frac{\sigma}{\sigma - 1} \frac{w}{(1 - \alpha_s) A_{si} K_{si}^{\alpha_s} L_{si}^{-\alpha_s}} \quad (8)$$

or equivalently

$$(1 - \tau_{Y_{si}}) P_{si} = \frac{\sigma}{\sigma - 1} \frac{(1 + \tau_{K_{si}}) R}{\alpha_s Y_{si} / K_{si}} = \frac{\sigma}{\sigma - 1} \frac{w}{(1 - \alpha_s) Y_{si} / L_{si}} \quad (9)$$

Note:

- $\tau_{Y_{si}}$ should be interpreted as any distortion that prevents the markup from reaching its desired level $\frac{\sigma}{\sigma - 1}$.
- It affects both K_{si} and L_{si} but does not change $\frac{K_{si}}{L_{si}}$ if we assume that $\tau_{Y_{si}}$ and $\tau_{K_{si}}$ are independent of each other.
- In the derivations of FOCs, I used repeatedly the property of Cobb-Douglas production function $\text{MPK} = \alpha \frac{Y}{K}$ and $\text{MPL} = (1 - \alpha) \frac{Y}{L}$. This simplifies the algebra by a lot, but at some cost (will see more in 701).
- Finally, I establish the relationship between firm-level variables (K_{si}, L_{si}, Y_{si}) and sector-level variables (K_s, L_s, Y_s) for future use. Using (6) to replace L_{si} with K_{si} gives

$$Y_{si} = A_{si} K_{si} \left(\frac{(1 + \tau_{K_{si}}) R}{w} \frac{1 - \alpha_s}{\alpha_s} \right)^{1 - \alpha_s}$$

Substituting this expression into (9) and replacing P_{si} with Y_{si} using the demand

function (4) gives

$$(1-\tau_{Ysi})P_s Y_s^{1/\sigma} \left(A_{si} K_{si} \left(\frac{(1+\tau_{Ksi})R}{w} \frac{1-\alpha_s}{\alpha_s} \right)^{1-\alpha_s} \right)^{-\sigma} = \frac{\sigma}{\sigma-1} \frac{(1+\tau_{Ksi})R}{\alpha_s A_{si} \left(\frac{(1+\tau_{Ksi})R}{w} \frac{1-\alpha_s}{\alpha_s} \right)^{1-\alpha_s}}$$

and therefore

$$K_{si} \propto (1-\tau_{Ysi})^\sigma A_{si}^{\sigma-1} (1+\tau_{Ksi})^{(1-\alpha_s)(\sigma-1)-\sigma}$$

Here I “drop” all the terms that are common across firms (i.e. model parameters and sectoral variables) because these terms will also show up in the aggregate capital K_s so that they will cancel out when we take K_{si}/K_s , and I keep only the firm-level variables. Hence, we know that

$$\frac{K_{si}}{K_s} = \frac{(1-\tau_{Ysi})^\sigma A_{si}^{\sigma-1} (1+\tau_{Ksi})^{(1-\alpha_s)(\sigma-1)-\sigma}}{\sum_i (1-\tau_{Ysi})^\sigma A_{si}^{\sigma-1} (1+\tau_{Ksi})^{(1-\alpha_s)(\sigma-1)-\sigma}} \quad (10)$$

Similarly, we get

$$\frac{L_{si}}{L_s} = \frac{(1-\tau_{Ysi})^\sigma A_{si}^{\sigma-1} (1+\tau_{Ksi})^{\alpha_s(1-\sigma)}}{\sum_i (1-\tau_{Ysi})^\sigma A_{si}^{\sigma-1} (1+\tau_{Ksi})^{\alpha_s(1-\sigma)}} \quad (11)$$

$$\frac{Y_{si}}{Y_s} = \frac{(1-\tau_{Ysi})^\sigma A_{si}^\sigma (1+\tau_{Ksi})^{(1-\alpha_s)\sigma-\sigma}}{\sum_i (1-\tau_{Ysi})^\sigma A_{si}^\sigma (1+\tau_{Ksi})^{(1-\alpha_s)\sigma-\sigma}} \quad (12)$$

$$\frac{P_{si}}{P_s} = \frac{(1-\tau_{Ysi})^{-1} A_{si}^{-1} (1+\tau_{Ksi})^{-(1-\alpha_s)+1}}{\sum_i (1-\tau_{Ysi})^{-1} A_{si}^{-1} (1+\tau_{Ksi})^{-(1-\alpha_s)+1}} \quad (13)$$

The takeaway from these equations is that the share of each firm in sectoral aggregates depends on how the firm-level productivity and the firm-level distortion compare to the average level in the sector.

From the last two equations, I can derive a ratio $\frac{P_{si}Y_{si}}{P_s Y_s}$: since

$$P_{si}Y_{si} \propto (1-\tau_{Ysi})^{\sigma-1} A_{si}^{\sigma-1} (1+\tau_{Ksi})^{\alpha_s(1-\sigma)}$$

hence

$$\frac{P_{si}Y_{si}}{P_s Y_s} = \frac{(1-\tau_{Ysi})^{\sigma-1} A_{si}^{\sigma-1} (1+\tau_{Ksi})^{\alpha_s(1-\sigma)}}{\sum_i (1-\tau_{Ysi})^{\sigma-1} A_{si}^{\sigma-1} (1+\tau_{Ksi})^{\alpha_s(1-\sigma)}} \quad (14)$$

This will be used later when we define sectoral MRPL and MRPK.

1.2 Productivity Measures

Before proceeding, I introduce several concepts that are useful for thinking about productivity. The pre-tax revenue of firm i is $P_{si}Y_{si}$. We can compute the partial derivatives of revenue with respect to capital and labor; these derivatives are called the “marginal revenue product of capital” and “marginal revenue product of labor,” respectively:

$$\begin{aligned}
 \text{MRPL}_{si} &\equiv \frac{\partial P_{si}Y_{si}}{\partial L_{si}} \\
 &= \frac{P_t Y_t^{1/\sigma} Y_{si}^{1-1/\sigma}}{\partial L_{si}} \\
 &= P_t Y_t^{1/\sigma} (1 - 1/\sigma) Y_{si}^{-1/\sigma} \frac{\partial Y_{si}}{L_{si}} \\
 &= \underbrace{P_t Y_t^{1/\sigma} Y_{si}^{-1/\sigma}}_{=P_{si}} (1 - 1/\sigma) (1 - \alpha_s) \frac{Y_{si}}{L_{si}} \\
 &= P_{si} \frac{\sigma - 1}{\sigma} (1 - \alpha_s) \frac{Y_{si}}{L_{si}} \\
 &= \frac{w}{1 - \tau_{Ysi}}
 \end{aligned} \tag{15}$$

Similarly

$$\begin{aligned}
 \text{MRPK}_{si} &\equiv \frac{\partial P_{si}Y_{si}}{\partial K_{si}} \\
 &= P_{si} \frac{\sigma - 1}{\sigma} \alpha_s \frac{Y_{si}}{K_{si}} \\
 &= \frac{(1 + \tau_{Ksi})R}{1 - \tau_{Ysi}}
 \end{aligned} \tag{16}$$

Note:

- If $\tau_{Ysi} = 0$ for every i , then MRPL_{si} is the same for all firms in industry s . In other words, there is no within-industry dispersion in MRPL. Hence, within-industry MRPL dispersion in the data indicates the presence of distortions. For this reason, the misallocation literature usually refers to dispersion in MRPL as “labor misallocation.”
- Similarly, dispersion in MRPK is referred to as “capital misallocation” in the literature, for exactly the same reason. The only difference is that now there is another contributor to the capital misallocation, which is the capital-specific distortion τ_{Ksi} . If $\tau_{Ysi} = 0$ and $\tau_{Ksi} = 0$, there would be no within-industry dispersion in MRPK.

Next, we go to the productivity measures. TFPQ is the Solow residual of the quantity

production function:

$$\text{TFPQ}_{si} = \frac{Y_{si}}{K_{si}^{\alpha_s} L_{si}^{1-\alpha_s}} = A_{si} \quad (17)$$

TFPR, on the other hand, is the Solow residual of the revenue function:

$$\text{TFPR}_{si} = \frac{P_{si} Y_{si}}{K_{si}^{\alpha_s} L_{si}^{1-\alpha_s}} = P_{si} A_{si} \quad (18)$$

Note:

- Since we allow firms to have different productivity, we should expect that there would be within-industry TFPQ dispersion regardless of the distortions.
- However, it is no longer the case for TFPR. We will illustrate this later.

The quantities defined thus far are at the firm level, and we can define their sector-level counterparts. Aggregate firm-level capital and labor to the sector level:

$$K_s = \sum_{i \in s} K_{si} \text{ and } L_s = \sum_{i \in s} L_{si}$$

Then we can define sectoral TFPQ as

$$\text{TFP}_s = \frac{Y_s}{K_s^{\alpha_s} L_s^{1-\alpha_s}} \quad (19)$$

and sectoral TFPR as

$$\text{TFPR}_s = \frac{P_s Y_s}{K_s^{\alpha_s} L_s^{1-\alpha_s}} \quad (20)$$

In the paper, TFPR_s has a bar over it, so I will use $\overline{\text{TFPR}}_s$ from now on.

Lastly, I can aggregate the sectoral output into the aggregate output

$$Y = \prod_s Y_s^{\theta_s} = \prod_s (\text{TFP}_s K_s^{\alpha_s} L_s^{1-\alpha_s})^{\theta_s} = \prod_s \text{TFP}_s^{\theta_s} \prod_s K_s^{\alpha_s \theta_s} \prod_s L_s^{(1-\alpha_s) \theta_s}$$

We define the non-input component in aggregate output as aggregate TFP:

$$\text{TFP} = \prod_s \text{TFP}_s^{\theta_s} \quad (21)$$

1.3 Relationship between Productivity Measures

TFPR vs. MRPL and MRPK

First, consider the relationship between total factor revenue productivity and the factor revenue products. Using the definition of TFPR and the penultimate lines in the definitions of MRPL and MRPK, we obtain

$$\text{TFPR}_{si} = \frac{\sigma}{\sigma - 1} \left(\frac{\text{MRPL}_{si}}{1 - \alpha_s} \right)^{1 - \alpha_s} \left(\frac{\text{MRPK}_{si}}{\alpha_s} \right)^{\alpha_s} = \frac{\sigma}{\sigma - 1} \frac{(1 + \tau_{Ksi})^{\alpha_s}}{1 - \tau_{Ysi}} \left(\frac{w}{1 - \alpha_s} \right)^{1 - \alpha_s} \left(\frac{R}{\alpha_s} \right)^{\alpha_s} \quad (22)$$

Note:

- Equation (22) shows that dispersion in TFPR comes only from the plant-specific distortions τ_{Ysi} and τ_{Ksi} ; it is independent of plant-level productivity dispersion.
- Hence, dispersion in TFPR is a better proxy than dispersion in TFPQ for the degree of misallocation in the economy. The smaller the variance of TFPR is, the more efficient is the economy.

Sectoral TFP vs firm-level TFP

Next, consider the relationship between sectoral TFP and firm-level productivity measures. Equation (15) in the paper is probably the most important equation for the empirical analysis:

$$\text{TFP}_s = \left(\sum_{i=1}^{M_s} \left(A_{si} \frac{\overline{\text{TFPR}}_s}{\text{TFPR}_{si}} \right)^{\sigma-1} \right)^{\frac{1}{\sigma-1}} \quad (23)$$

where

$$\overline{\text{TFPR}}_s \equiv \left(\frac{\sigma}{\sigma - 1} \right) \left(\frac{\overline{\text{MRPK}}_s}{\alpha_s} \right)^{\alpha_s} \left(\frac{\overline{\text{MRPL}}_s}{1 - \alpha_s} \right)^{1 - \alpha_s}$$

and

$$\begin{aligned} \overline{\text{MRPK}}_s &\equiv \frac{R}{\sum_i \frac{1 - \tau_{Ysi}}{1 + \tau_{Ksi}} \frac{P_{si} Y_{si}}{P_s Y_s}} \\ \overline{\text{MRPL}}_s &\equiv \frac{w}{\sum_i (1 - \tau_{Ysi}) \frac{P_{si} Y_{si}}{P_s Y_s}} \end{aligned}$$

Let's derive this equation. First, let's derive the expression for $\overline{\text{TFPR}}_s$. To do that, we first compute the denominator, $K_s^{\alpha_s} L_s^{1 - \alpha_s}$. Recall the FOC (9) implies that

$$K_{si} = \left(1 - \frac{1}{\sigma} \right) \frac{1 - \tau_{Ysi}}{1 + \tau_{Ksi}} \frac{\alpha_s}{R} P_{si} Y_{si}$$

and

$$L_{si} = \left(1 - \frac{1}{\sigma} \right) \frac{1 - \tau_{Ysi}}{1} \frac{1 - \alpha_s}{w} P_{si} Y_{si}$$

Hence, sector-level capital and labor can be aggregated as

$$K_s = \sum_i K_{si} = \left(1 - \frac{1}{\sigma}\right) \frac{\alpha_s}{R} \sum_i \frac{1 - \tau_{Ysi}}{1 + \tau_{Ksi}} P_{si} Y_{si}$$

and

$$L_s = \sum_i L_{si} = \left(1 - \frac{1}{\sigma}\right) \frac{1 - \alpha_s}{w} \sum_i \frac{1 - \tau_{Ysi}}{1} P_{si} Y_{si}$$

Hence

$$K_s^{\alpha_s} L_s^{1-\alpha_s} = \left(1 - \frac{1}{\sigma}\right) \left(\frac{\alpha_s}{R}\right)^{\alpha_s} \left(\frac{1 - \alpha_s}{w}\right)^{1-\alpha_s} \left(\sum_i \frac{1 - \tau_{Ysi}}{1 + \tau_{Ksi}} P_{si} Y_{si}\right)^{\alpha_s} \left(\sum_i \frac{1 - \tau_{Ysi}}{1} P_{si} Y_{si}\right)^{1-\alpha_s}$$

and therefore

$$\begin{aligned} \overline{\text{TFPR}}_s &= \frac{P_s Y_s}{K_s^{\alpha_s} L_s^{1-\alpha_s}} \\ &= \frac{P_s Y_s}{\left(1 - \frac{1}{\sigma}\right) \left(\frac{\alpha_s}{R}\right)^{\alpha_s} \left(\frac{1 - \alpha_s}{w}\right)^{1-\alpha_s} \left(\sum_i \frac{1 - \tau_{Ysi}}{1 + \tau_{Ksi}} P_{si} Y_{si}\right)^{\alpha_s} \left(\sum_i \frac{1 - \tau_{Ysi}}{1} P_{si} Y_{si}\right)^{1-\alpha_s}} \\ &= \frac{\sigma}{\sigma - 1} \left(\frac{R}{\alpha_s \sum_i \frac{1 - \tau_{Ysi}}{1 + \tau_{Ksi}} \frac{P_{si} Y_{si}}{P_s Y_s}}\right)^{\alpha_s} \left(\frac{w}{(1 - \alpha_s) \sum_i \frac{1 - \tau_{Ysi}}{1} \frac{P_{si} Y_{si}}{P_s Y_s}}\right)^{1-\alpha_s} \\ &= \left(\frac{\sigma}{\sigma - 1}\right) \left(\frac{\overline{\text{MRPK}}_s}{\alpha_s}\right)^{\alpha_s} \left(\frac{\overline{\text{MRPL}}_s}{1 - \alpha_s}\right)^{1-\alpha_s} \end{aligned} \quad (24)$$

Next, we move from $\overline{\text{TFPR}}_s$ to TFP_s using the price index

$$P_s = \left(\sum_i P_{si}^{1-\sigma}\right)^{\frac{1}{1-\sigma}} \quad (25)$$

Since

$$P_{si} = \frac{\text{TFPR}_{si}}{A_{si}}$$

and

$$P_s = \frac{\overline{\text{TFPR}}_s}{\text{TFP}_s}$$

Plug them into the price index:

$$\frac{\overline{\text{TFPR}}_s}{\text{TFP}_s} = \left(\sum_i \left(\frac{\text{TFPR}_{si}}{A_{si}}\right)^{1-\sigma}\right)^{\frac{1}{1-\sigma}}$$

Rearranging terms gives

$$\text{TFP}_s = \left(\sum_i A_{si}^{\sigma-1} \frac{\overline{\text{TFPR}}_s^{\sigma-1}}{\text{TFPR}_{si}^{\sigma-1}} \right)^{\frac{1}{\sigma-1}} \quad (26)$$

Note:

- As discussed above, if there were no distortions, then $\tau_{Ysi} = \tau_{Ksi} = 0$ and $\text{TFPR}_{si} = \overline{\text{TFPR}}_s$. We can then define the efficient benchmark for sectoral TFP as

$$\text{TFP}_s^* = \left(\sum_i A_{si}^{\sigma-1} \right)^{\frac{1}{\sigma-1}} \quad (27)$$

It is simply an unweighted average of firm-level productivity A_{si} .

- If we shut off capital-specific distortion, i.e. $\tau_{Ksi} = 0$, we can simplify (26) by combining the definitions of $\overline{\text{TFPR}}_s$, TFPR_{si} and the weights $\frac{P_{si}Y_{si}}{P_sY_s}$ in equation (14):

$$\text{TFP}_s = \frac{(\sum_i A_{si}^{\sigma-1} (1 - \tau_{Ysi})^{\sigma-1})^{\frac{\sigma}{\sigma-1}}}{\sum_i (1 - \tau_{Ysi})^\sigma A_{si}^{\sigma-1}} \quad (28)$$

- We are interested in how the sectoral TFP_s deviates from the efficient benchmark TFP_s^* . The ratio

$$\frac{\text{TFP}_s}{\text{TFP}_s^*}$$

helps us quantify the percentage loss of sectoral TFP from its efficient benchmark. When there are a lot of firms in the sector (so that $\sum_i$ can be replaced by integral $\int_0^1$), capital-specific distortion is shut off ($\tau_{Ksi} = 0$) and the joint distribution of $\log(1 - \tau_{Ysi})$ and $\log A_{si}$ is a Gaussian distribution

$$\begin{bmatrix} \log(1 - \tau_{Ysi}) \\ \log A_{si} \end{bmatrix} \sim N \left(\begin{bmatrix} \bar{\tau} \\ \bar{a} \end{bmatrix}, \begin{bmatrix} \sigma_\tau^2 & \sigma_{a\tau} \\ \sigma_{a\tau} & \sigma_a^2 \end{bmatrix} \right)$$

then we can show from (28) that

$$\begin{aligned}
\text{TFP}_s &\approx \frac{\mathbb{E} [A_{si}^{\sigma-1} (1 - \tau_{Ysi})^{\sigma-1}]^{\frac{\sigma}{\sigma-1}}}{\mathbb{E} [(1 - \tau_{Ysi})^\sigma A_{si}^{\sigma-1}]} \\
&= \frac{\mathbb{E} [\exp((\sigma-1) \log A_{si} + (\sigma-1) \log(1 - \tau_{Ysi}))]^{\frac{\sigma}{\sigma-1}}}{\mathbb{E} [\exp(\sigma \log(1 - \tau_{Ysi}) + (\sigma-1) \log A_{si})]} \\
&= \frac{\exp((\sigma-1)\bar{a} + (\sigma-1)\bar{\tau} + \frac{1}{2}(\sigma-1)^2(\sigma_\tau^2 + \sigma_a^2 + 2\sigma_{a\tau}))^{\frac{\sigma}{\sigma-1}}}{\exp(\sigma\bar{\tau} + (\sigma-1)\bar{a} + \frac{1}{2}(\sigma^2\sigma_\tau^2 + (\sigma-1)^2\sigma_a^2 + 2\sigma(\sigma-1)\sigma_{a\tau}))} \\
&= \exp\left(\bar{a} - \frac{1}{2}\sigma\sigma_\tau^2 + \frac{1}{2}(\sigma-1)\sigma_a^2\right)
\end{aligned}$$

and the efficient TFP_s^* is

$$\text{TFP}_s^* = \exp\left(\bar{a} + \frac{1}{2}(\sigma-1)\sigma_a^2\right)$$

Hence

$$\log \text{TFP}_s - \log \text{TFP}_s^* = -\frac{1}{2}\sigma\sigma_\tau^2 \quad (29)$$

This is equation (16) in the paper, and it is the main equation for the quantitative analysis in the paper. Note that this does not hold for the case where $\tau_{Ksi} = 0$. Equation (29) gives us the percentage loss of sectoral TFP from its efficient benchmark. The TFP loss gets larger when σ is larger and when σ_τ^2 is larger (i.e. when the distortion is more dispersed).

Note that from the second line to the third line in the derivation of TFP_s above, we have used the property of log normal distribution: if $\log X \sim N(\mu, \sigma^2)$, then $\mathbb{E}[X] = \exp(\mu + \frac{1}{2}\sigma^2)$.

Aggregate TFP, Sectoral TFP and Firm-level TFP

Lastly, we define allocative efficiency at aggregate level by the ratio of aggregate TFP and the desired TFP (in which each sector is efficient with $\text{TFP}_s = \text{TFP}_s^*$):

$$\text{AE} \equiv \frac{\prod_s \text{TFP}_s^{\theta_s}}{\prod_s (\text{TFP}_s^*)^{\theta_s}} = \prod_s \left(\sum_i \left(\frac{A_{si}}{\text{TFP}_s^*} \frac{\overline{\text{TFPR}}_s}{\text{TFPR}_{si}} \right)^{\sigma-1} \right)^{\frac{\theta_s}{\sigma-1}} \quad (30)$$

Note:

- Equation (30) helps quantify the aggregate TFP loss of the economy and it is the equation (20) in the paper. Note that the left-hand side of the equation in the paper

is wrong - aggregate TFP loss is not the same as aggregate output loss $Y/Y_{\text{efficient}}$.

- If there is no TFPR dispersion, i.e., $\text{TFPR}_{si} = \overline{\text{TFPR}_s}$, allocative efficiency equals 1. TFPR dispersion leads to $\text{AE} < 1$.
- The percentage improvement in aggregate TFP from removing TFPR dispersion can therefore be calculated as $1/\text{AE} - 1$.

2 Empirics (Very Brief)

From the U.S. Census, obtain the following variables:

- industry;
- labor compensation (wL_{si});
- value added ($P_{si}Y_{si}$);
- export revenues;
- capital stock K_{si} .

One can also work with more accessible data containing these variables, such as Compustat data for U.S. public firms.

Set

- rental price of capital $R = 0.1$; can do so because the efficiency gain does not depend on R ;
- elasticity of substitution $\sigma = 3$ (low, compared to the literature, but also gives us a conservative quantification of misallocation);
- elasticity of output with respect to capital in each sector, α_s , is set to be 1 minus labor share in the corresponding sector, which can be retrieved from NBER Productivity Database (Census/ASM). There are better ways in the literature to uncover α_s from production function estimation (e.g. Olley and Pakes, Levinsohn and Petrin, ACF... more in 701).

Hence, we can infer the wedges or distortions from the data:

$$1 + \tau_{K_{si}} = \frac{\alpha_s}{1 - \alpha_s} \frac{wL_{si}}{RK_{si}} \quad (31)$$

$$1 - \tau_{Y_{si}} = \frac{\sigma}{\sigma - 1} \frac{wL_{si} + RK_{si}}{P_{si}Y_{si}} \quad (32)$$

$$A_{si} = \frac{Y_{si}}{K_{si}^{\alpha_s} L_{si}^{1-\alpha_s}} = \kappa_s \frac{(P_{si}Y_{si})^{\frac{\sigma}{\sigma-1}}}{K_{si}^{\alpha_s} (wL_{si})^{1-\alpha_s}} \quad (33)$$

where

$$\kappa_s \equiv w^{1-\alpha_s} (P_s Y_s)^{-\frac{\sigma}{\sigma-1}} Y_s$$

The last equation recovers TFPQ. Since Y_{si} is not directly observable in the data, we need to transform it into the observables. Need to take advantage of the fact that

$$P_s Y_s^{1/\sigma} = P_{si} Y_{si}^{1/\sigma}$$

one can easily check that the last equality is correct.

Again, κ_s is not directly observable, but it does not contribute to the distortions. Hence, the authors normalize $\kappa_s = 1$ for all industries.

One can perform a similar analysis for China and India. The empirical section of the paper presents the resulting implications.